

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2 ASSESSMENT TOOL 1 – Scenario-Based Requirement Analysis

Course Code:	CSA10 / CSA1003	Course Title:	Software Engineering
CO Assessed:	CO2	Assessment:	Assessment Tool 2 – Scenario-Based Requirement Analysis
Weightage:	20%	Total Marks:	25
CO2 Statement:	<i>Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills</i>		
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Date:	03-08-2026	Duration:	60 minutes

Code of Conduct

I K V CHARAN TEJA(192424167) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: K V CHARAN TEJA_____

ACADEMIC PROJECT, ALLOCATION MONITORING AND EVALUATION MANAGEMENT SYSTEM

Question

Based on your **assigned problem statement**, perform an **Architecture Design Analysis** and prepare an architecture design report by answering the following questions:

Project Title

Academic Project Allocation, Monitoring and Evaluation Management System

1. Introduction

The **Academic Project Allocation, Monitoring and Evaluation Management System** is a web-based application developed to automate the process of project allocation, monitoring, evaluation, and reporting in educational institutions. The system helps students, faculty members, project coordinators, HODs, administrators, and management manage academic projects efficiently.

The existing system mainly depends on manual processes and project allocation based on CGPA, which does not consider students' interests or technical skills. The proposed system provides a transparent, efficient, and digital platform for managing the complete project lifecycle.

2. Study and Analyze the Scenario

Business Domain

Education Management System

Existing Challenges

- Manual project allocation process.
 - Projects allocated only based on CGPA.
 - Student interests and technical skills are ignored.
 - Poor communication among stakeholders.
 - Difficult to monitor project progress.
 - Manual evaluation process.
 - Project records are difficult to maintain.
 - Time-consuming report generation.
-

Stakeholders

- Students
 - Faculty Guides
 - Project Coordinators
 - Head of Department (HOD)
 - Administrator
 - External Examiner
 - College Management
-

System Objectives

- Allocate projects according to students' interests and skills.
 - Improve project quality.
 - Digitally monitor project progress.
 - Maintain project records.
 - Ensure fair evaluation.
 - Improve communication.
 - Reduce manual work.
 - Generate reports automatically.
-

3. Functional Requirements

FR1 User Registration

The system shall allow users to register.

FR2 User Login

The system shall authenticate users based on username and password.

FR3 Student Profile Management

Students can update

- Personal Details
 - CGPA
 - Technical Skills
 - Preferred Domain
 - Resume
-

FR4 Interest Selection

Students can choose

- Domain
 - Programming Languages
 - Technical Skills
 - Project Preferences
-

FR5 Project Allocation

The Project Coordinator shall allocate projects based on

- Skills
 - Interests
 - Domain
 - Availability
-

FR6 Faculty Guide Assignment

Assign suitable faculty guide.

FR7 Project Monitoring

Faculty shall monitor

- Weekly Progress
 - Milestones
 - Attendance
 - Submission Status
-

FR8 Evaluation

Faculty and External Examiner shall evaluate

- Internal Marks
 - External Marks
 - Viva
 - Documentation
-

FR9 Feedback

Faculty can provide

- Suggestions
 - Comments
 - Improvement Points
-

FR10 Report Generation

Generate

- Student Reports
 - Faculty Reports
 - Allocation Reports
 - Evaluation Reports
 - Department Reports
-

FR11 User Management

Administrator manages

- Users
 - Departments
 - Projects
 - Roles
-

4. Non-Functional Requirements

Performance

- Response time should be less than 3 seconds.
 - Support multiple users simultaneously.
-

Security

- Secure Login
 - Password Encryption
 - Role-Based Access
 - Data Protection
-

Reliability

- Minimum system failures
 - Accurate data storage
 - Data integrity
-

Usability

- User-friendly interface
 - Easy navigation
 - Responsive design
-

Scalability

The system should support increasing numbers of

- Students
 - Faculty
 - Departments
 - Projects
-

Maintainability

- Easy software updates
 - Easy bug fixing
 - Modular architecture
-

Availability

- System available 24×7
 - Minimum downtime
-

Backup and Recovery

- Daily backup
 - Automatic recovery
-

5. Actors and Use Cases**Primary Actors**

- Student
 - Faculty Guide
 - Project Coordinator
 - Administrator
-

Secondary Actors

- HOD
 - External Examiner
 - Management
-

Student Use Cases

- Register
-

- Login
 - Manage Profile
 - Select Skills
 - Select Interests
 - View Allocated Project
 - Upload Progress
 - View Feedback
 - View Marks
-

Faculty Guide Use Cases

- Login
 - View Assigned Students
 - Monitor Progress
 - Review Reports
 - Evaluate Project
 - Provide Feedback
-

Project Coordinator Use Cases

- Manage Projects
 - Allocate Projects
 - Assign Faculty
 - Generate Reports
-

Administrator Use Cases

- Manage Users
 - Manage Departments
 - Manage Projects
 - View Reports
-

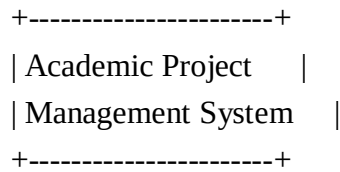
HOD Use Cases

- View Department Reports
 - Monitor Progress
 - Approve Projects
-

External Examiner Use Cases

- Login
 - Evaluate Project
 - Submit Marks
-

6. Use Case Diagram



Student -----> Register

Student -----> Login

Student -----> Manage Profile

Student -----> Select Skills

Student -----> Upload Progress

Faculty -----> Monitor Progress

Faculty -----> Evaluate Project

Faculty -----> Feedback

Coordinator -----> Allocate Project

Coordinator -----> Assign Faculty

Administrator -----> Manage Users

Administrator -----> Generate Reports

HOD -----> View Reports

External Examiner -----> Evaluate Project

7. Data Requirements

The system stores

Student Data

- Student ID
 - Name
 - Register Number
 - Email
 - Phone
 - CGPA
 - Skills
 - Interests
-

Faculty Data

- Faculty ID
 - Name
 - Department
 - Specialization
-

Project Data

- Project ID
 - Project Title
 - Domain
 - Description
 - Status
-

Allocation Data

- Student ID
 - Faculty ID
 - Project ID
-

Progress Reports

- Week Number
 - Progress
 - Remarks
-

Evaluation Data

- Internal Marks
 - External Marks
 - Viva Marks
-

Feedback Data

- Comments
 - Suggestions
-

8. External Interface Requirements

User Interface

- Web Browser
 - Mobile Responsive
-

Hardware Interface

- Computer
 - Laptop
 - Smartphone
-

Software Interface

- MySQL Database
 - REST API
-

- Email API
 - SMS Gateway
 - Cloud Storage
-

Communication Interface

- HTTP
 - HTTPS
 - REST API
 - SQL
 - SMTP
-

SOFTWARE REQUIREMENT SPECIFICATION (SRS)

Academic Project Allocation, Monitoring and Evaluation Management System

1. INTRODUCTION

The Academic Project Allocation, Monitoring and Evaluation Management System is a web-based application developed to automate the process of project allocation, monitoring, evaluation, and reporting in educational institutions.

2. PROBLEM DESCRIPTION

In the existing system, projects are allocated based on CGPA without considering students' interests, technical skills or preferred domains. This reduces motivation and learning opportunities. Manual monitoring and evaluation are time-consuming, error-prone and lack transparency.

3. SYSTEM OBJECTIVES

- Allocate projects based on students' interests and skills.
- Improve practical learning and project quality.
- Monitor project progress efficiently.
- Maintain project records digitally.
- Make project evaluation fair and transparent.
- Improve communication among all stakeholders.
- Reduce manual work and save time.

4. SCOPE OF THE SYSTEM

The system is for colleges to manage the complete lifecycle of academic projects. It includes user management, project allocation, faculty assignment, progress tracking, evaluation, feedback and report generation. It supports Students, Faculty, Project Coordinators, HOD, External Examiners, Administrator and Management.

5. FUNCTIONAL REQUIREMENTS

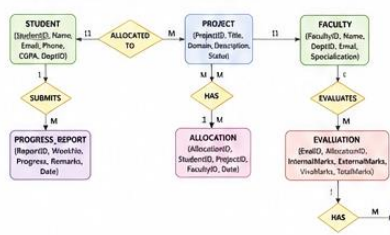
FR1	User Registration – System shall allow users to register.
FR2	User Login – System shall authenticate users.
FR3	Student Profile Management – Students can update profile, CGPA, skills, interests and resume.
FR4	Interest & Skill Selection – Students select domain, skills and project preferences.
FR5	Project Allocation – Coordinator allocates projects based on skills, interests and availability.
FR6	Faculty Guide Assignment – System assigns suitable faculty guide.
FR7	Project Progress Tracking – Faculty monitors weekly progress, milestones and submission.
FR8	Evaluation – Internal and external evaluation by faculty and external examiner.
FR9	Feedback Management – Faculty provides feedback and suggestions.
FR10	Report Generation – Generate student, faculty, allocation and evaluation reports.
FR11	User & Project Management – Administrator manages users, projects, departments and roles.

6. NON-FUNCTIONAL REQUIREMENTS

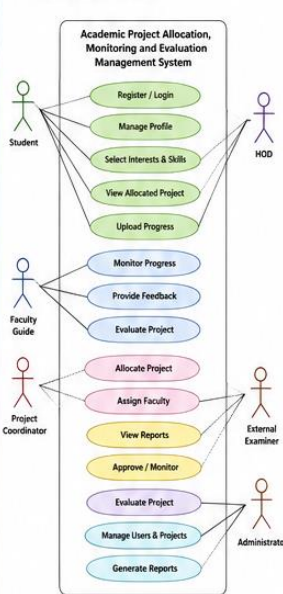
NFR	DESCRIPTION
NFR1	Performance – Response time < 3 seconds; support multiple users.
NFR2	Security – Secure login, data encryption, role-based access.
NFR3	Reliability – Minimum failures, accurate and consistent data.
NFR4	Usability – User-friendly, easy navigation, responsive UI.
NFR5	Scalability – Support increasing users, departments and projects.
NFR6	Maintainability – Easy to modify, update and maintain.
NFR7	Availability – System available 24x7 with minimal downtime.
NFR8	Backup & Recovery – Daily backup and data recovery.

7. ACTORS

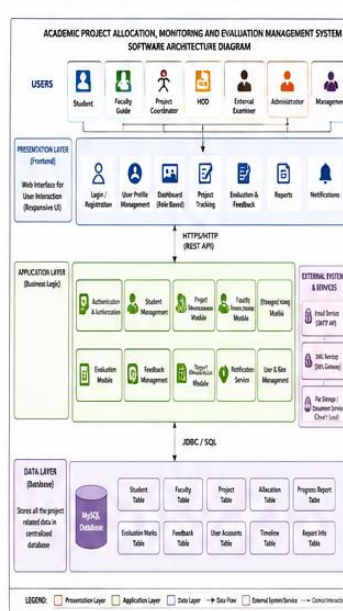
Actor Type	Actors
Primary Actors	Student, Faculty Guide, Project Coordinator, Administrator
Secondary Actors	HOD, External Examiner, Management



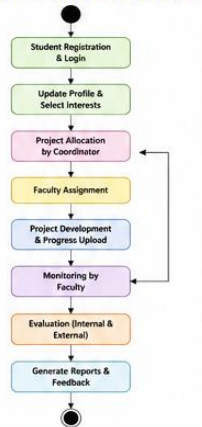
8. USE CASE DIAGRAM



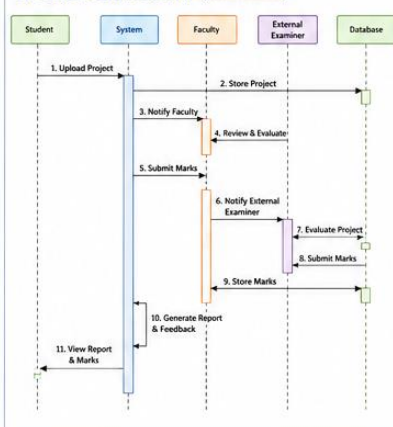
9. SOFTWARE ARCHITECTURE DIAGRAM



10. ACTIVITY DIAGRAM (Project Lifecycle)

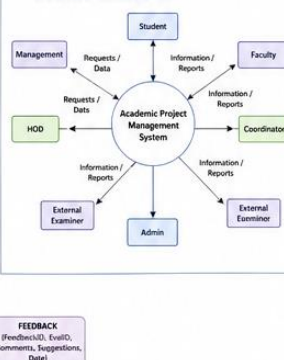


11. SEQUENCE DIAGRAM (Project Evaluation)

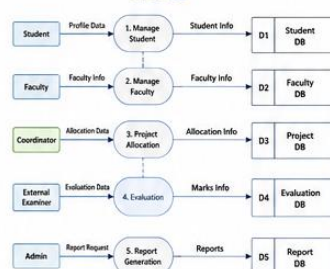


13. DATA FLOW DIAGRAM

DFD LEVEL 0 (Context Diagram)



DFD LEVEL 1



14. ASSUMPTIONS

- Internet connection is available.
- Users have valid institutional credentials.
- Users will provide accurate information.
- Faculty and students will follow the process.
- Data backup will be performed regularly.

15. CONSTRAINTS

- Limited budget and resources.
- Time constraint (semester timeline).
- Dependence on server and internet.
- Compliance with institutional policies.
- Data privacy and legal regulations.

16. REQUIREMENT VALIDATION

- ✓ All stakeholders identified.
- ✓ Functional and non-functional requirements covered.
- ✓ Security, performance and scalability addressed.
- ✓ Data requirements and interfaces specified.
- ✓ No ambiguity, redundancy or inconsistency.
- ✓ Requirements are clear, complete and testable.

17. SUGGESTED IMPROVEMENTS

- AI-based project recommendation.
- Plagiarism detection integration.
- Online viva and presentation module.
- Mobile application for easy access.
- Analytics dashboard and predictions.

9. System Features

- Secure Login
 - Role-Based Access
 - Student Dashboard
 - Faculty Dashboard
 - HOD Dashboard
 - Admin Dashboard
 - Project Dashboard
 - Project Allocation
 - Faculty Assignment
 - Progress Tracking
 - Evaluation
 - Feedback
 - Report Generation
 - Notifications
-

10. Assumptions

- Internet connection is available.
 - Users have valid college credentials.
 - Database server is available.
 - Faculty update project status regularly.
 - Students upload reports on time.
 - Authorized users access the system.
-

11. Constraints

Technical Constraints

- Internet dependency
 - Server availability
 - Browser compatibility
-

Operational Constraints

- User training required
 - Institutional approval required
-

Economic Constraints

- Limited development budget
 - Infrastructure cost
-

Legal Constraints

- Student data privacy
 - Institutional policies
-

Time Constraints

- Project completion within semester
 - Timely testing and deployment
-

12. Validation of Requirement Completeness and Consistency

Requirement Validation

- ✓ All stakeholders identified.
 - ✓ Functional requirements completely covered.
 - ✓ Non-functional requirements included.
 - ✓ Security requirements addressed.
 - ✓ Data requirements specified.
 - ✓ External interfaces defined.
 - ✓ No duplicate requirements.
 - ✓ Requirements are consistent.
 - ✓ Requirements are understandable.
 - ✓ Future scalability considered.
-

13. Suggested Improvements

- AI-based project recommendation.
 - Plagiarism detection.
 - Online Viva Module.
 - Mobile Application.
 - Cloud Backup.
 - LMS Integration.
 - Email & SMS Notifications.
 - Analytics Dashboard.
-

Rubrics for Calculation

Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Scenario Understanding and Problem Analysis (30%)	Demonstrates thorough understanding of the scenario, stakeholders, objectives, and business context with clear analysis.	Demonstrates good understanding with minor omissions.	Basic understanding with limited analysis.	Poor understanding; major aspects missing or incorrect.
Functional and Non-Functional Requirement Identification (25%)	Clearly identifies comprehensive, accurate, and well-classified functional and non-functional requirements.	Identifies most requirements with minor classification errors.	Identifies only basic requirements; several omissions.	Requirements are incomplete, inaccurate, or poorly classified.
Actor Identification and Use Case Modelling (20%)	Correctly identifies all relevant actors and use cases; use case diagram is complete, accurate, and follows UML standards.	Most actors and use cases identified; diagram has minor errors.	Limited actors/use cases identified; diagram partially correct.	Actors/use cases missing or incorrect; diagram absent or inaccurate.
Software Requirement Specification (SRS) Documentation (15%)	SRS is well-structured, complete, professionally organized, and includes all required sections.	SRS is organized with minor missing sections or formatting issues.	SRS includes only essential sections with limited detail.	SRS is incomplete, poorly organized, or lacks essential components.
Assumptions, Constraints, and Requirement Validation (10%)	Clearly identifies realistic assumptions and constraints; validates requirements for completeness, consistency, and ambiguity with appropriate justification.	Identifies most assumptions and constraints; validation is adequate.	Limited assumptions/constraints; validation is superficial.	Assumptions, constraints, and validation are missing or incorrect.

Criteria	Mark
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Scenario Understanding and Problem Analysis (30%)	/5
Functional and Non-Functional Requirement Identification (25%)	/5
Actor Identification and Use Case Modelling (20%)	/5
Software Requirement Specification (SRS) Documentation (15%)	/5
Assumptions, Constraints, and Requirement Validation (10%)	/5
TOTAL	/25